

9.

Ans: C

All geostationary orbits has the same orbital period T as Earth ie 24 hours. They must orbit at the same radius from center of Earth.

Thus, angular velocity, $\omega = \frac{2\pi}{T}$, is the same and centripetal acceleration $a = r\omega^2$ is also the same.

Since mass of satellite, m may be different, $KE = \frac{1}{2}mv^2 = \frac{1}{2}m(r\omega)^2$ may be different.

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Ans: D